

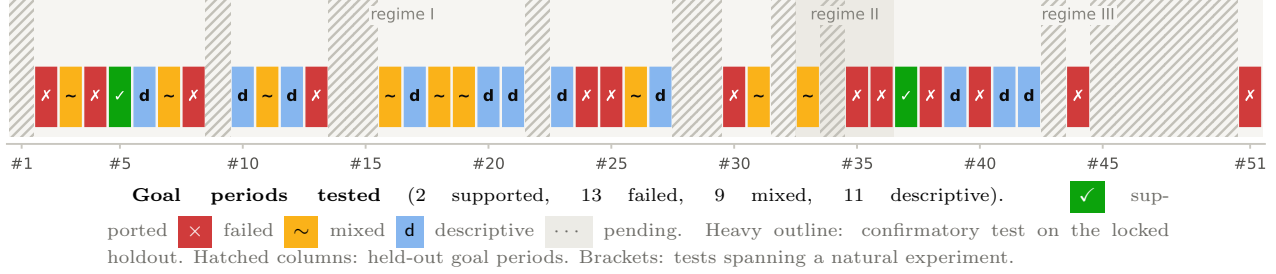
H03: Swarm activity is self-exciting, and its criticality is set by the goal's coupling mode

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

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Bottom line. The swarm is clearly subcritical ($n \approx 0.2\text{--}0.4$) and its self-excitation does not depend on the goal's coupling mode; only a small, fast, reply-like cross-triggering survives its null.

level: hypothesis · pipeline checks 71% · updated 2026-10-04



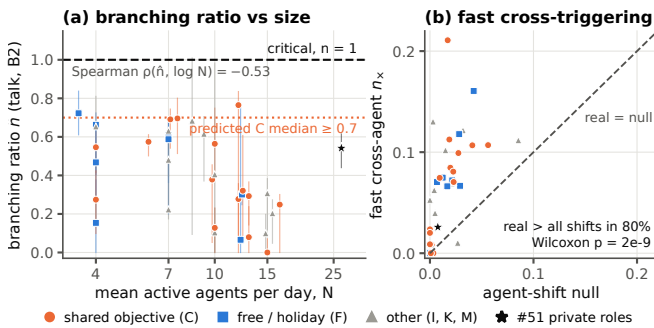
WHAT HAPPENED

Hawkes fits to agent talk events in 35 non-holdout goal periods (282 days), each fitted separately with a schedule baseline. Predicted: shared-objective weeks run near critical (median $\hat{n} \geq 0.7$) and above free weeks, and #51 drifts toward $n = 1$ as its roster grows. All failed. Median $\hat{n}$ is 0.41 (range 0–0.76), bracketed at 0.2–0.4 by the baseline choice. Shared-objective weeks gave 0.32 vs 0.53 for free weeks ($p = 0.75$); $\hat{n}$ falls with swarm size ($\rho = -0.53$), and in #51 as its roster grows ($\rho = -0.64$). A constant baseline would have “confirmed” criticality ($\hat{n} \approx 0.99$), and pooled $\hat{n}$ cannot be separated from 10–30-min rate modulation. The robust positive: fast cross-agent triggering, $n_{\times} \approx 0.07$ at 10–30 s lags, about 4× an agent-shift null ($p = 2 \times 10^{-9}$). Humans and the nudger directly drive about 1.5% of events. No holdout run yet.

PHYSICAL MODEL

Model 09 (Hawkes self-exciting point process). Events: agent messages; one realization per village day. $\mu_B(t)$: schedule baseline (per-day level $\times$ 30-min shape, plus a kickoff bump); g_{exo} : human and nudger input; ϕ : triggering kernel; n : branching ratio, the mean number of events each event directly triggers (critical at $n = 1$; $\langle s \rangle$: mean cascade size). Multivariate fit: λ_a , agent a 's intensity, with own and cross kernels; $\bar{m}$: active agents; $w_m^{\times}$: cross weight at timescale τ_m .

$$\begin{aligned}\lambda(t) &= \mu_B(t) + \sum_{x < t} g_{\text{exo}}(t - x) + \sum_{t_j < t} \phi(t - t_j) \\ \phi(\tau) &= n\beta e^{-\beta\tau}, \quad \langle s \rangle = 1/(1 - n) \\ \lambda_a(t) &= c_a\mu(t) + \sum_{t_j \in a} \phi_{\text{self}}(t - t_j) + \sum_{t_j \notin a} \phi_{\times}(t - t_j) \\ n_{\times} &= (\bar{m} - 1) \sum_{\tau_m \leq 300\text{s}} w_m^{\times}\end{aligned}$$



Observables. (a) Branching ratio per goal period against swarm size, by coupling mode, with day-bootstrap 95% CIs: none is near critical, shared-objective periods sit no higher than free ones, and $\hat{n}$ falls with size. (b) Fast cross-agent excitation against an agent-shift null that breaks cross-agent timing: nearly every period sits above the diagonal, the one component that beats its strongest null.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

A negative that matters a little: this swarm is nowhere near a runaway cascade, and neither more agents nor a shared goal pushes it there. Anyone worried about message storms gets mild reassurance, but activity is set mostly by the scheduler and roster, not by social contagion. The one robust number (about 0.07 extra messages from others within half a minute of each message) may just be agents waking when a message lands: a scaffold property. A naive constant-baseline Hawkes fit would have reported near-criticality. Practically: one pit-fall and one small coupling constant.

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